

ORACLE DBA PRACTICE

An End-to-End Guide for Beginners

Practical, Step-by-Step Database Administration

```
SQL> SELECT instance_name, status  
2 FROM v$instance;
```

INSTANCE_NAME	STATUS
DB01	OPEN

```
SQL> █
```



Backup



Recovery



Monitoring



Administration



Presented by S³ Academy

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Oracle Apps DBA Support | Corporate & College Training

ORACLE DBA

PRACTICAL NOTES

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HYDERABAD

Preface

Oracle database administration is a discipline built on technical understanding, careful execution, and sound judgment. Knowing a command is a starting point. Understanding when to use it, what it changes, and how to verify its outcome is the foundation of dependable administration.

Oracle DBA Practice: An End-to-End Guide for Beginners is recommended as a learning and lab-practice resource by **Mr. Syed Shabaz, Hyderabad**, who has **30 years of experience in Oracle Database and Oracle Applications DBA**. This compilation brings together practical commands, lab exercises, and administration notes to help learners connect database concepts with hands-on practice.

The book covers installation, instance startup and shutdown, parameter files, tablespace management, user administration, backup and recovery, RMAN, Data Pump, ASM, Data Guard, and performance tuning. Its examples include **Oracle 11g and 12c environments**, reflecting the technologies used in the underlying practice material. Readers should consult the relevant Oracle documentation when adapting these examples to other releases.

The value of practical learning lies in understanding the complete procedure, beyond reproducing its commands. Before beginning an exercise, identify its objective, confirm the environment, and understand the required privileges and prerequisites. During execution, observe the results. Afterwards, verify the outcome through appropriate database views, logs, and checks.

A command that completes successfully does not, by itself, confirm that the objective has been achieved. A backup requires recovery validation. A configuration change requires assessment of its effect. A performance problem requires investigation supported by evidence. These distinctions help learners develop the judgment needed to administer databases responsibly.

Readers are encouraged to maintain their own practice records: the task performed, commands executed, outputs observed, errors encountered, and steps taken to resolve them. Such records strengthen understanding and provide a useful reference for future troubleshooting.

For Learning and Lab Practice Only

This book is intended strictly for education and practice in a dedicated, isolated lab environment. Its commands, scripts, and procedures are not intended for direct use on production or live databases.

Do not copy or execute examples on a live system without fully understanding their purpose, prerequisites, dependencies, and potential impact. Database names, usernames, credentials, directory paths, and storage settings are illustrative and must be adapted consistently to your lab setup.

Review and test each procedure in a suitable non-production environment. Compile scripts where applicable and validate their results. **Successful compilation or execution alone does not establish that a procedure is safe for production.**

Any production activity requires a separately reviewed and approved procedure, appropriate authorization, verified backups where relevant, and a tested rollback or recovery plan. Incorrect execution may cause data loss, service interruption, or unintended configuration changes.

A Note to Readers

Although this material has been compiled with care, inadvertent typographical, editorial, or technical errors may remain. If you notice an error or an unclear explanation, please bring it to our attention so it can be reviewed and corrected. **Your constructive feedback will help make future editions more accurate and useful.**

If an instruction appears incomplete, inconsistent, or unsuitable for your environment, pause and verify it before proceeding. Consult the official documentation for your Oracle release and seek experienced guidance when needed.

For those pursuing Oracle Applications DBA responsibilities, database knowledge provides an essential foundation. Administration of Oracle E-Business Suite also involves the application tier, operating system, patching, cloning, and supporting services. This book's database exercises should therefore be complemented by focused application administration study and supervised experience.

This compilation is intended to be a working companion throughout the early stages of learning. Approach each exercise with curiosity and discipline. Investigate unexpected results, revisit the concepts behind the procedure, and develop the habit of verifying rather than assuming.

Technical confidence grows through understanding, repeatable practice, and reliable evidence. We hope this book helps students, aspiring DBAs, and IT professionals build those habits and progress toward thoughtful, capable database administration.

Recommended for Learning & Lab Practice by

Mr. Syed Shabaz - Hyderabad

30 Years of Experience in Oracle Database & Oracle Applications DBA

Presented by S³ Academy

Empowering Minds. Shaping Futures.

Preface - Hinglish

Oracle database administration ek aisi discipline hai jo technical samajh, saavdhani se kaam karne aur sahi faisle lene par aadharit hai. Kisi command ko jaanna shuruaat hai. Use kab chalana hai, woh kya badalti hai aur uska result kaise verify karna hai - yeh samajh bharosemand administration ki buniyad hai.

Oracle DBA Practice: An End-to-End Guide for Beginners ko **Mr. Syed Shabaz, Hyderabad**, jinke paas **Oracle Database aur Oracle Applications DBA mein 30 years ka experience** hai, learning aur lab practice ke liye recommend karte hain. Is compilation mein practical commands, lab exercises aur administration notes ko saath rakha gaya hai, taaki learners database concepts ko hands-on practice se jod saken.

Is book mein installation, instance startup aur shutdown, parameter files, tablespace management, user administration, backup aur recovery, RMAN, Data Pump, ASM, Data Guard aur performance tuning shamil hain. Examples mein **Oracle 11g aur 12c environments** ka use hai, jo original practice material mein istemal hui technologies ko darshata hai. Dusre releases ke liye examples adapt karte samay sambandhit Oracle documentation zaroor dekhein.

Practical learning ka mahatva sirf commands dobara chalane mein nahi, balki poori procedure ko samajhne mein hai. Exercise shuru karne se pehle uska objective, environment, required privileges aur prerequisites check karein. Execution ke dauran results dekhein. Uske baad suitable database views, logs aur checks se outcome verify karein.

Command ka successfully complete hona apne aap mein yeh confirm nahi karta ki objective achieve ho gaya hai. Backup ke liye recovery validation zaroori hai. Configuration change ke effect ko assess karna chahiye. Performance problem ki investigation evidence par aadharit honi chahiye. Yeh samajh learners ko databases ki zimmedari se administration karne mein madad karti hai.

Apne practice records maintain karein: kiya gaya task, chalayi gayi commands, dekhe gaye outputs, saamne aaye errors aur unhe resolve karne ke steps. Aise records understanding ko mazboot banate hain aur future troubleshooting ke liye useful reference dete hain.

Sirf Learning aur Lab Practice ke Liye

Yeh book strictly education aur dedicated, isolated lab environment mein practice ke liye hai. Iski commands, scripts aur procedures production ya live databases par seedhe use karne ke liye nahi hain.

Examples ka purpose, prerequisites, dependencies aur possible impact poori tarah samjhe bina unhe live system par copy ya execute na karein. Database names, usernames, credentials, directory paths aur storage settings illustrative hain. Inhe apne lab setup ke hisaab se consistently adapt karein.

Har procedure ko suitable non-production environment mein review aur test karein. Jahan applicable ho, scripts compile karein aur results validate karein. **Sirf successful compilation ya execution se koi procedure production ke liye safe prove nahi hoti.**

Production activity ke liye alag se reviewed aur approved procedure, proper authorization, jahan relevant ho verified backups, aur tested rollback ya recovery plan zaroori hai. Galat execution se data loss, service interruption ya unintended configuration changes ho sakte hain.

Readers ke Liye Ek Note

Is material ko saavdhani se compile kiya gaya hai, phir bhi anjaane mein typographical, editorial ya technical errors reh sakte hain. Agar koi error ya unclear explanation mile, to humein batayein, taaki use review aur correct kiya ja sake. **Aapka constructive feedback future editions ko aur accurate aur useful banane mein madad karega.**

Agar koi instruction incomplete, inconsistent ya aapke environment ke liye unsuitable lage, to aage badhne se pehle ruk kar verify karein. Apne Oracle release ki official documentation dekhein aur zaroorat par experienced guidance lein.

Oracle Applications DBA responsibilities seekhne walon ke liye database knowledge ek zaroori foundation hai. Oracle E-Business Suite administration mein application tier, operating system, patching, cloning aur supporting services bhi shamil hain. Isliye book ki database exercises ke saath focused application administration study aur supervised experience bhi zaroori hai.

Yeh compilation learning ke early stages mein ek working companion ke roop mein taiyar ki gayi hai. Har exercise ko curiosity aur discipline ke saath karein. Unexpected results ki investigation karein, procedure ke peeche ke concepts dobara samjhein aur assume karne ke bajay verify karne ki aadat banayein.

Technical confidence samajh, repeatable practice aur reliable evidence se badhta hai. Humein umeed hai ki yeh book students, aspiring DBAs aur IT professionals ko yeh habits develop karne aur soch-samajhkar database administration karne mein madad karegi.

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Index

Full-book page references. This free sample contains pages 1-15 only.

Topics	Pages
ORACLE INSTALLATION 11G	9-20
Startup stages	21-22
Front Page (How to connect to the DB Server Machine)	23-27
Manual Database Creation Script	28-30
Parameter File Management (pfile & spfile)	31-39
Tablespace Management - I	40-61
Tablespace Management - II	62-76
CONTROL FILE AND REDOLOG FILE MANAGEMENT	77-90
USER MANAGEMENT-I	91-113
USER MANAGEMENT - II	114-123
Backup & Recovery (User Managed)	124-135
Oracle Networking-I	136-155
Oracle Networking -II	156-159
Cloning	160-170
Recovery Manager - I	171-180
Recovery Manager - II	181-204
Recovery Manager- 3	205-241

Index - Continued

Full-book page references. This free sample contains pages 1-15 only.

Topics	Pages
Recovery Manager-IV	242-272
Datapump Utilities	273-301
Logminer and SQL Loader	302-308
Flashback Feature	309-330
DB SERVER SOFTWARE PATCH MANAGEMENT	331-354
Upgradation and Migration	355-366
ORACLE INSTALLATION 12C	367-376
Automatic Storage Management	377-399
DATABASE MIGRATION FROM NON-ASM TO ASM	400-408
Oracle Data Guard (An Oracle Enterprise Edition Feature)	409-417
DATA GUARD MODES AND FAILOVER, SWITCHOVER- II	418-430
Performance Tuning	431-441
AWR REPORT (Automatic Workload Repository)	442-462
INDEXES	463-470
Troubleshooting	471-480
REAL APPLICATION TESTING	481-484
Shortcut Keys	485

ORACLE INSTALLATION 11G

System requirement (For Practice)

For oracle 10g:-

Physical memory (RAM)	512 MB recommended
Virtual memory	Double the amount of RAM
Temp disk space	Hard disk space 1.5 GB
Video adapter	256 colors
CPU Processor	200 MHz minimum

For oracle 11g:-

Physical memory (RAM)	1GB minimum
Virtual memory	Double the amount of RAM
Temp disk space	Hard disk space 4.76 GB
Video adapter	256 colors
CPU Processor	550 MHz minimum

Create a usergroup and osuser

```
[root@S3_ACADEMY ~] # groupadd dba
```

```
[root@S3_ACADEMY ~]# df -h
```

Filesystem	Size	Used	Avail	Use%	Mounted on
/dev/sda1	15G	6.6G	7.0G	49%	/
/dev/sda5	5.7G	141M	5.3G	3%	/tmp
/dev/sda2	70G	180M	66G	1%	/u01
Tmpfs	2.0G	0	2.0G	0%	/dev/shm

```
[root@S3_ACADEMY ~] # useradd -g dba -d /u01/home -m oracle
```

-g →to specify group name

-d →to specify user's home directory

-m →to create home directory if it is not created

```
[root@S3_ACADEMY ~]# chown oracle:dba /u01/home
```

```
[root@S3_ACADEMY ~]# chmod 775 /u01/home
```

```
[root@S3_ACADEMY ~]# xhost +
```

access control disabled, clients can connect from any host

```
[root@S3_ACADEMY ~]# su - oracle
```

```
[oracle@S3_ACADEMY ~]$ whoami
```

oracle

```
[oracle@S3_ACADEMY ~]$ pwd
```

/u01/home

```
[oracle@S3_ACADEMY ~]$
```

HOW TO SHARE S/W

step 1: click on vm

step 2: select settings

step 3 select options

step 4 : select shared folders enabled

step 5 : click on always enabled

click on add
click on next
step 6 : browse (select s/w location)
click on next
step 7 : click on finish
step 8 : click on ok

THEN

```
[oracle@S3_ACADEMY ~]$ cd /mnt
```

```
[oracle@S3_ACADEMY mnt]$ ls
```

hgfs

```
[oracle@S3_ACADEMY mnt]$ cd hgfs
```

```
[oracle@S3_ACADEMY hgfs]$ ls
```

11gr2p3

```
[oracle@S3_ACADEMY hgfs]$ cd 11gr2p3/
```

```
[oracle@S3_ACADEMY 11gr2p3]$ ls
```

doc readme.html rpm sshsetup welcome.html

install response runInstaller stage

```
[oracle@S3_ACADEMY 11gr2p3]$ ./runInstaller
```

Starting Oracle Universal Installer...

Checking Temp space: must be greater than 120 MB. Actual 5372 MB Passed

Checking swap space: must be greater than 150 MB. Actual 8001 MB Passed

Checking monitor: must be configured to display at least 256 colors. Actual 16777216 Passed

Preparing to launch Oracle Universal Installer from /tmp/OraInstall2016-09-04_11-18-55PM. Please wait ...

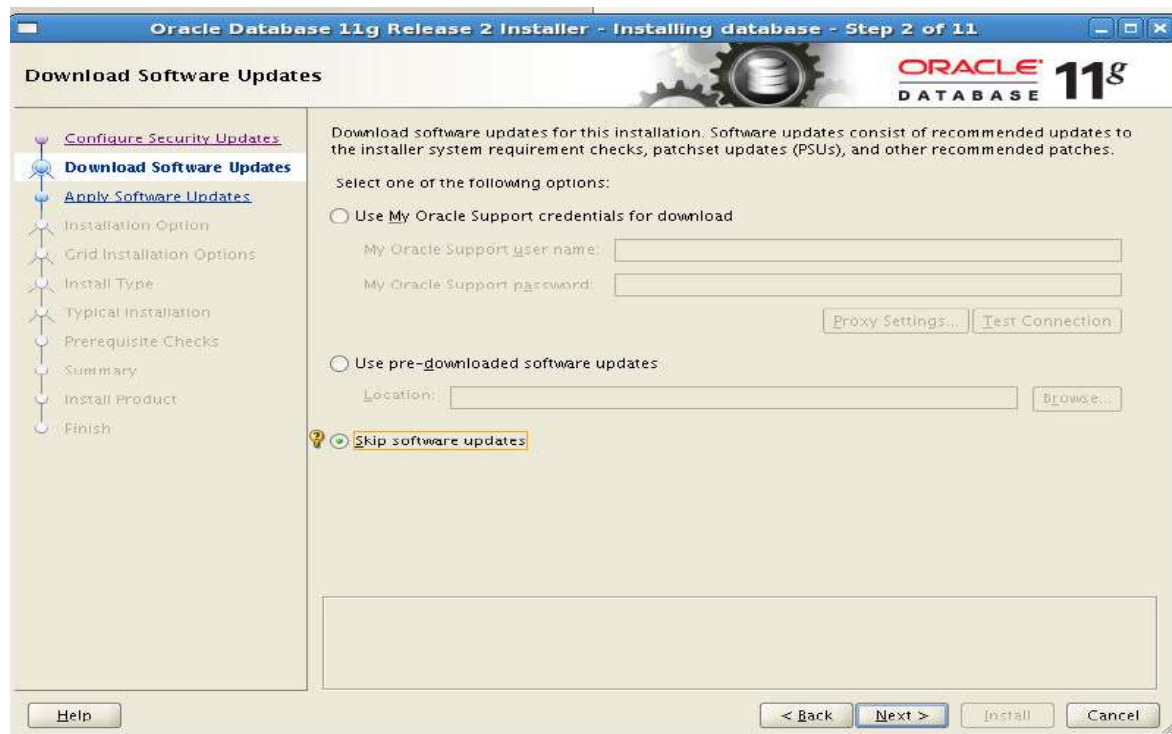
Step :1 uncheck (I wish to receive updates via my oracle support)click next



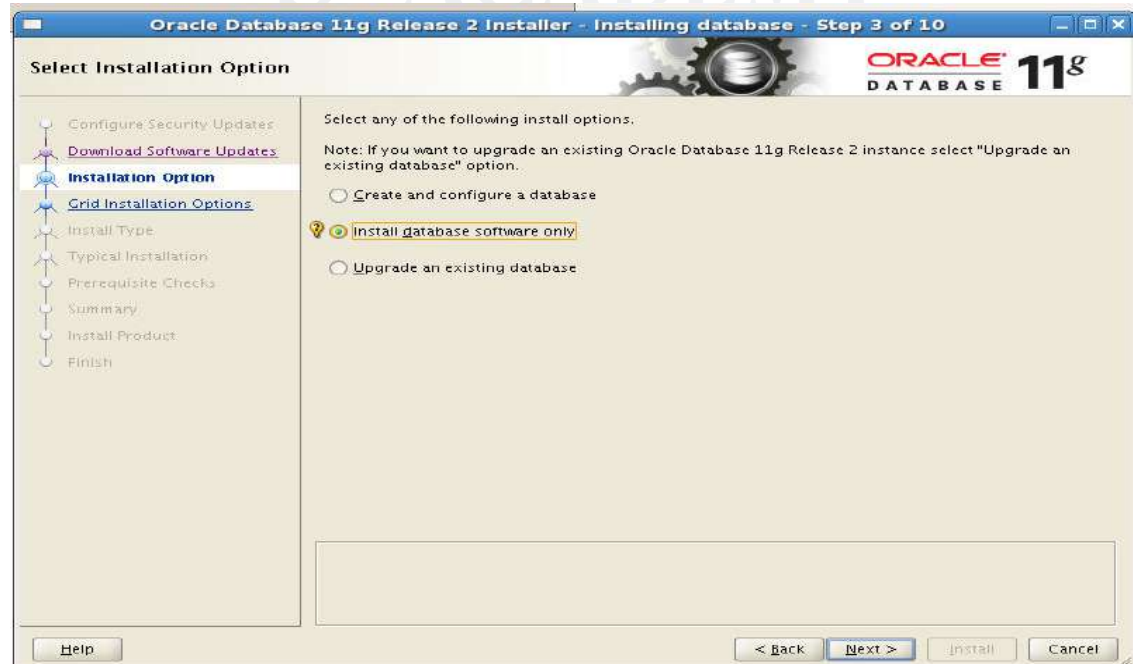
Step :2 click on yes.



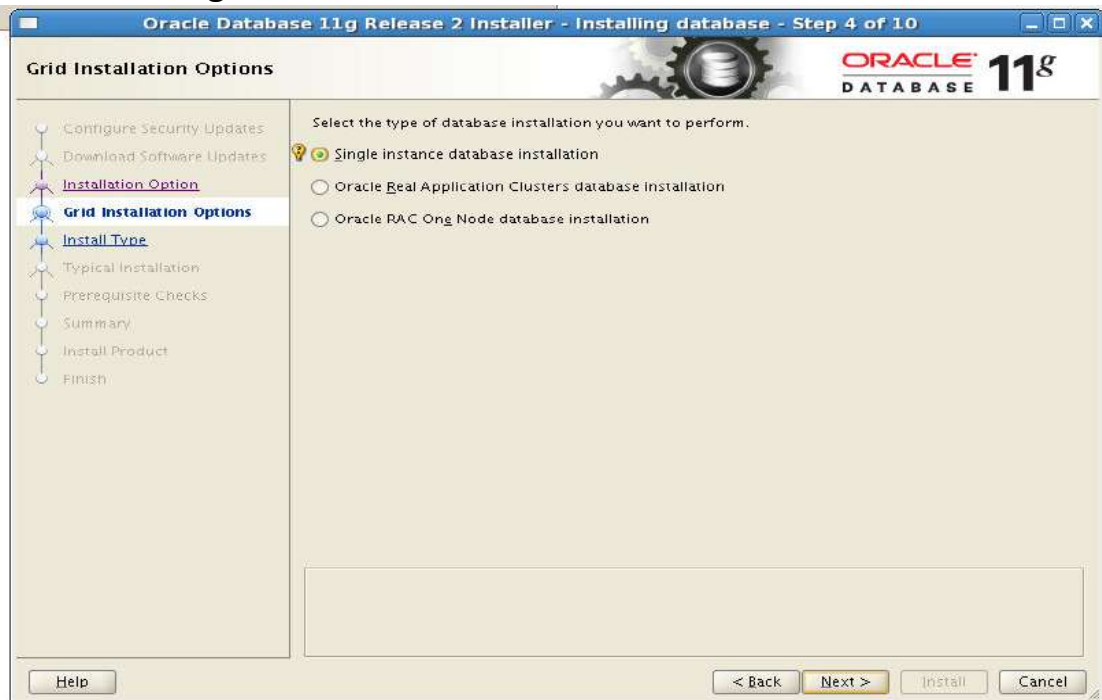
Step :3 select on skip software click next



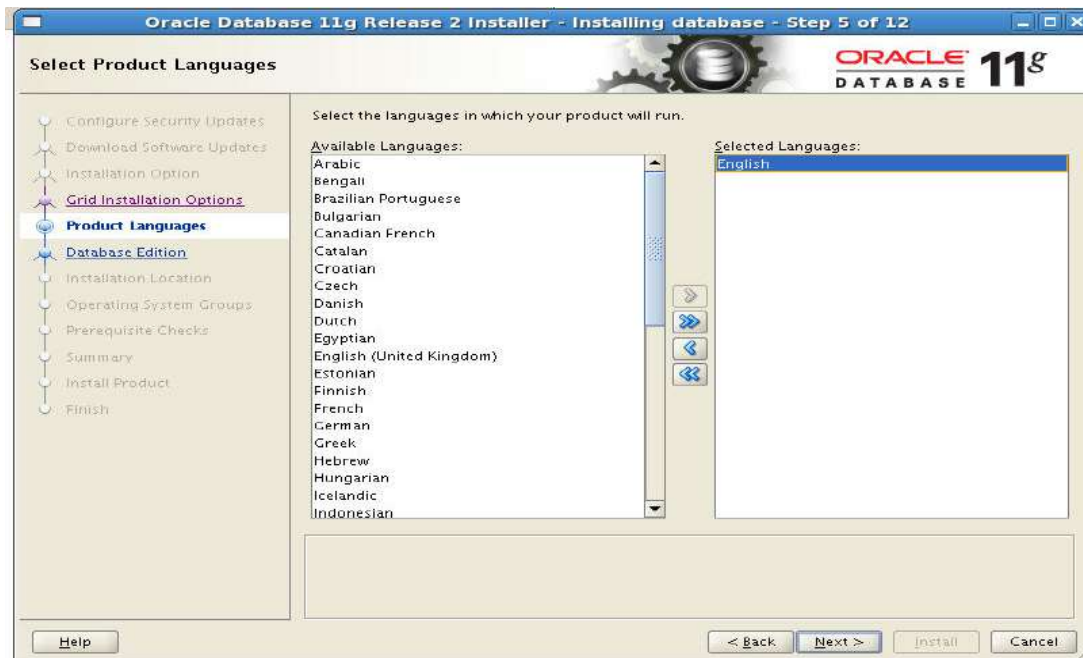
Step :4 select install database s/w only



Step :5 select single instance database installation click next



Step :6 select language (ex :English) click next



Step :7 select enterprise edition click next



Step :8 give oracle base and s/w location

(ex : /u01/home/app/product/11.2.0/dbhome_1)

Click next

Specify Installation Location

Specify a path to place all Oracle software and configuration-related files installed by this installation owner. This location is the Oracle base directory for the installation owner.

Oracle Base: /u01/home/app

Specify a location for storing Oracle database software files separate from database configuration files in the Oracle base directory. This software directory is the Oracle database home directory.

Software Location: /u01/home/app/product/11.2.0/dbhome_1

Step :9 give inventory directory (ex: /u01/home/12c/orainventory)

Click next

Create Inventory

You are starting your first installation on this host. Specify a directory for installation files. This directory is called the "inventory directory". The installer automatically sets up subdirectories for each product to contain inventory data. The subdirectory for each product typically requires 150 kilobytes of disk space.

Inventory Directory: /u01/home/orainventory

Specify an operating system group whose members have write permission to the inventory directory (orainventory).

orainventory Group Name: dba